GEOTWINAI

AI-Powered Digital Twin for Smart Cities

Nagpur Urban Planning & Decision Support Report

Study Area	Nagpur, Maharashtra
Satellite	Sentinel-2 Level-2A
Satellite Source	Microsoft Planetary Computer
Spatial Resolution	10 metres
Scene Date	2026-05-12
Generated By	GeoTwinAI Automated Report Generator

GeoTwinAI integrates geospatial infrastructure datasets, satellite remote sensing, environmental indices, flood analysis, heatmap analysis and machine learning to support smart-city planning and decision-making.

1. Executive Summary

GeoTwinAI is an AI-powered geospatial decision-support platform developed for urban planning and smart-city analysis in Nagpur, Maharashtra. The platform integrates infrastructure datasets, Sentinel-2 satellite observations, environmental indices, flood-risk analysis, heatmap analysis and machine-learning components. The selected Sentinel-2 scene has a reported cloud cover of 0.00% and a scene date of 2026-05-12. The environmental analysis includes NDVI, NDBI and NDWI. Infrastructure analysis includes buildings, hospitals, schools, parks, roads and water bodies. The platform additionally incorporates flood-risk and urban heatmap analysis to support environmental risk assessment. The final workflow connects GIS visualization, machine learning, automated data processing and Power BI analytics.

2. Project Objectives

• Develop an AI-powered Digital Twin framework for Nagpur.

• Integrate satellite and GIS datasets.

• Analyse vegetation and environmental conditions.

• Analyse built-up and water-related characteristics.

• Map important urban infrastructure.

• Analyse flood-risk conditions.

• Identify heat-related urban hotspots.

• Create machine-learning-based urban priority analysis.

• Provide interactive GIS visualization.

• Provide analytical dashboards through Power BI.

• Support data-driven urban planning decisions.

• Create an extensible framework for future data updates.

3. Data Sources

Dataset	Purpose
Buildings	Urban building and infrastructure analysis
Hospitals	Healthcare infrastructure mapping
Schools	Educational infrastructure mapping
Parks	Green and open-space analysis
Roads	Transportation network analysis
Water Bodies	Water-resource analysis
Sentinel-2	Satellite and environmental analysis
All Sentinel-2 Bands	Multi-spectral analysis
NDVI / NDBI / NDWI	Environmental and urban indicators
LULC	Land-use and land-cover analysis
Flood Risk	Flood vulnerability analysis

4. Urban Infrastructure Statistics

Infrastructure Type	Records
Buildings	46052
Hospitals	412
Schools	75
Parks	142
Roads	110566
Water Bodies	0

5. Sentinel-2 Satellite Analysis

The satellite analysis uses Sentinel-2 Level-2A imagery obtained through Microsoft Planetary Computer. The selected scene date is 2026-05-12, with a reported cloud cover of 0.00%. A common 10-metre spatial reference grid is used for integrated multi-band analysis. The processed dataset includes B01, B02, B03, B04, B05, B06, B07, B08, B8A, B09, B11 and B12.

6. Spectral Indices

Index	Purpose	Mean
NDVI	Vegetation condition	0.1392
NDBI	Built-up area indication	0.0804
NDWI	Water-related analysis	-0.1982

7. Sentinel-2 All-Band Analysis

The all-band Sentinel-2 dataset preserves spectral information required for environmental and urban analysis. Processed bands include: B01 - Coastal/Aerosol B02 - Blue B03 - Green B04 - Red B05 - Vegetation Red Edge B06 - Vegetation Red Edge B07 - Vegetation Red Edge B08 - Near Infrared B8A - Narrow Near Infrared B09 - Water Vapour B11 - Short-Wave Infrared B12 - Short-Wave Infrared These bands support vegetation, water, soil, built-up and land-cover analysis.

8. Machine Learning Analysis

The machine-learning component of GeoTwinAI is designed to classify urban areas according to planning priority. The feature-engineering stage combines infrastructure and environmental attributes into a machine-learning dataset. The trained model can generate urban priority predictions such as High, Medium and Low. These predictions can support identification of areas requiring additional infrastructure or environmental attention. Model performance should be evaluated using appropriate classification metrics such as accuracy, precision, recall and F1-score.

9. GeoTwinAI Workflow

1. Data collection

2. Data cleaning

3. Satellite processing

4. Spectral index generation

5. Feature engineering

6. Machine learning

7. Prediction

8. Interactive GIS visualization

9. Power BI dashboard

10. Urban planning decision support

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8. Land Use, Built-up Area and Green Cover

GeoTwinAI incorporates land-use and land-cover analysis to understand the spatial composition of the study area. Built-up analysis provides an indication of urbanized areas, while green-cover analysis supports identification of vegetation-rich and vegetation-deficient regions. LULC analysis can support planning decisions involving urban expansion, open spaces, vegetation conservation and land management.